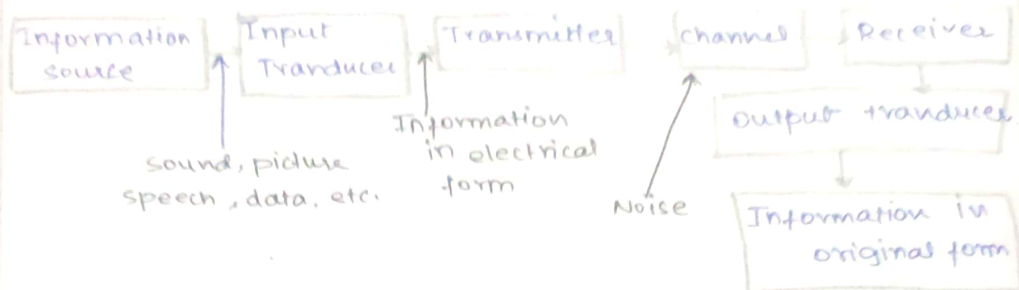


QUESTION-1

- (1) Draw the block diagram of a communication system and
 a) briefly explain the function of each block.



BLOCK DIAGRAM OF COMMUNICATION SYSTEM -

1. INFORMATION SOURCE :- The function of information source is to produce required message signal which has to be transmitted.
2. INPUT TRANSDUCER :- It is used to convert the original message signal into a time varying electrical signal.
3. TRANSMITTER :- The function of transmitter is to protect electrical signals from different aspects.
 Example: For radio broadband signal, the electrical signal obtained from sound signals, is processed to restricted range of audio frequency and is then amplified.
 Modulation is the main function of it.
4. CHANNEL :- The function of channel is to provide a physical connection between the transmitter and receiver. There are two types of channel point to point and broadcasting channel.
5. RECEIVER :- The main function of receiver is to reproduce the message signal in electrical signal form from the distorted received signal.

DESTINATION :- It is the final stage which is used to convert an electrical message signal into its original form.

(b) What is Baseband Transmission?

The message signal generated from the information source is known as baseband signal. The baseband signal may be combination of two or more message signals. If the baseband signal is transmitted directly then it is known as baseband transmission.

(c) Explain the need of modulation in communication system.

We need modulation for the following reasons:-

(1) Practicality of Antenna:-

For efficient radiation and reception the transmitting and receiving antennas must have length quarter to wavelength of the frequency used. Clearly it is impractical to install an antenna of such height.

Hence height of Antenna may reduce through mean of modulation technique and effective radiation and reception is achieved.

(2) To remove interference:-

To keep various Radio stations transmitting same sound signal of 20Hz to 20kHz range it is necessary to translate or shift them to different portions of electromagnetic spectrum. Thus each station is allocated a band of frequency. This also overcomes the drawback of poor radiation efficiency of low frequency.

QUESTION - 2

CSE214059

What is meant by the term Amplitude Modulation?

Amplitude modulation may be defined as the process in which amplitude of the carrier wave is varied according to instantaneous value of the modulating or baseband signal.

(b) Define Modulation Index for A.M wave.

Modulation Index (MI) symbolic form (m_a) is defined as the measure of extent of amplitude variation about an unmodulated maximum carrier.

$$m_a = \frac{x(t)_{\max}}{\text{Max carrier amp.}}$$

(c) Derive an expression for single tone Amplitude Modulated wave.

Mathematical Expression:-

let us consider a single tone modulating signal as

$$x(t) = V_m \cos \omega_m t \quad \text{---(1)} \quad \left| \omega_m \rightarrow \text{signal frequ.} \right.$$

let the carrier signal be,

$$c(t) = A \cos \omega_c t \quad \text{---(2)}$$

We know that general expression for AM signal is,

$$s(t) = [A + x(t)] \cos \omega_c t$$

$$\text{or } s(t) = A \cos \omega_c t + x(t) \cos \omega_c t$$

putting the value of $x(t)$ in equation (3) we get,

$$s(t) = A \cos \omega_c t + V_m \cos \omega_m t \cos \omega_c t$$

$$= A \left[1 + \frac{V_m}{A} \cos \omega_m t \right] \cos \omega_c t \quad \text{---(4)}$$

(4)

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But we know that for AM the modulation index m_a is,

$$m_a = \frac{|x(t)|_{\max}}{A}$$

where $|x(t)|_{\max}$ denotes the maximum amplitude of modulated signal and A is maximum amplitude of carrier signal.

In this case we have,

$$|x(t)|_{\max} = V_m$$

$$\therefore m_a = \frac{V_m}{A}$$

Now putting the value of (m_a) in equation (4) we get

$$s(t) = A [1 + m_a \cos \omega_m t] \cos \omega_c t \quad \text{--- (5)}$$

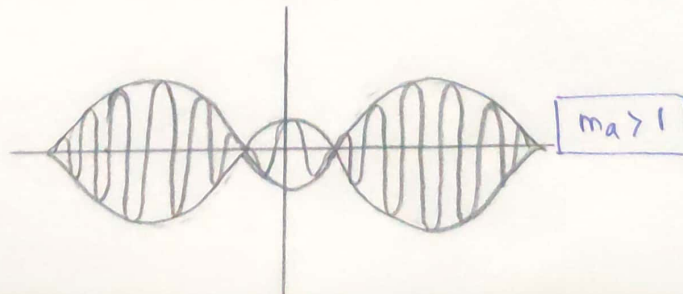
Hence the desired expression for single tone modulated signal is,

$$s(t) = A [1 + m_a \cos \omega_m t] \cos \omega_c t$$

- (d) Draw the waveform of an over modulated AM wave and write the condition for over Modulation.

OVER-MODULATION :

Over Modulated AM wave : When modulation index is greater than 1 i.e. $m_a = \frac{|x(t)|_{\max}}{A} > 1$, i.e. amplitude of message signal is greater than Amplitude of Carrier signal.



Question 3:-

(a) What is Carson's rule?

Carson's Rule provides a thumb formula to calculate the bandwidth of single tone WBFM.

According to this rule FM, BW is given as twice the sum of its frequency deviation and highest modulating signal frequency.

$$BW = 2(\Delta\omega + \omega_m) = 2\left(1 + \frac{\Delta\omega}{\omega_m}\right)\omega_m$$

$$BW = 2\omega_m(1 + m_f)$$

(b) Explain the generation of narrowband FM signals with suitable block diagram.

We know the general equation for FM wave in the phasor form is given by,

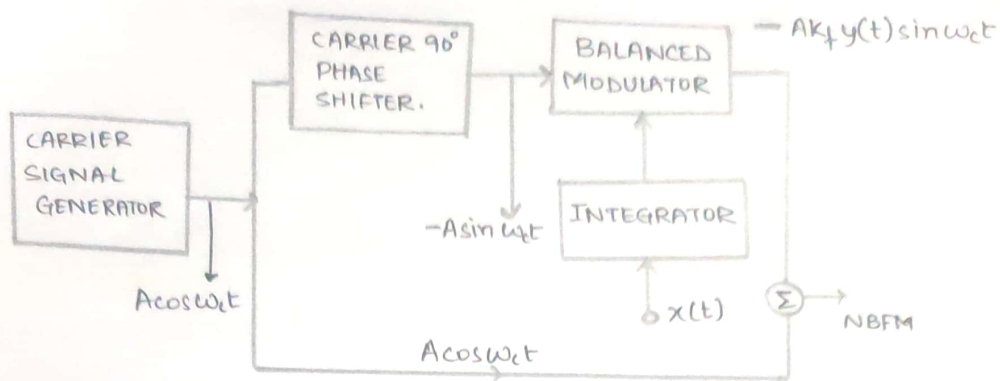
$$c_{FM} = Ae^{j(\omega_c t + k_f y(t))} \text{ or,}$$

$$s(t) = A \cos \omega_c t - A k_f y(t) \sin \omega_c t \quad \text{--- (1)}$$

The above equation provide an idea to generate narrowband FM.

These means that the sideband terms are generated by a balanced modulator as in a DSB-SC system, and then the carrier term is added to the sideband terms. Based upon the above equation we can show that method of generation of narrowband FM.

DIAGRAM →



NARROWBAND FM GENERATOR.

- (c) A baseband or modulating signal $x(t) = 5 \cos(2\pi \times 1000t)$ angle modulated a carrier $A \cos \omega_c t$.

i) Determine the modulation index and bandwidth and modulation index of FM if Modulating frequency is reduced to 5 KHz. Assume $k_f = 15 \text{ KHz/Volt}$.

GIVEN:

$$x(t) = 5 \cos(2\pi \times 1000t) \quad [\text{Modulating signal}]$$

comparing with actual formula, of Modulating signal
 $x(t) = V_m \cos \omega_m t$.

$$\therefore V_m = 5 \text{ V}$$

$$\omega_m = 2\pi \times 1000 = 2\pi f_m$$

$$f_m = 1000 \text{ Hz}$$

$$k_f = 15 \times 10^3 \text{ Hz}$$

$$(i) \text{ Modulating index } (m_f) = \frac{A_m k_f}{f_m} = \frac{5 \times 15 \times 10^3}{10^3} = 75$$

$$\text{Bandwidth (BW)} = 2 f_m [1 + m_f] = 2 \times 1000 [1 + 75] = 2 [75 + 1] 1000 = 152 \times 10^3 \text{ Hz or } 152 \text{ kHz.}$$

(ii) Now $f_m = 5 \text{ kHz}$.

$$\therefore \text{modulating index } (m_f) = \frac{A_m \times k_f}{f_m} = \frac{5 \times 15}{5} = 15$$

$$\text{Band width (BW)} = 2(m_f + 1)f_m = 2(15 + 1) \times 5 = 160 \text{ kHz}.$$

$$\text{change in } \Delta m_f = 35 - 15 = 60 \quad \text{and} \quad \Delta BW = 160 - 152 = 8 \text{ kHz}$$

QUESTION - 4

(a) State sampling theorem.

A bandwidth limited continuous time signal of finite energy may be completely represented in its samples, and recovered back if the sampling frequency is

$f_s \geq 2f_m$. Hence f_s is the sampling frequency and f_m is the maximum frequency present in the signal.

(b) What is aliasing? How is it prevented?

Aliasing is an effect that causes different signals to become indistinguishable (or aliased of one another) when sampled.

To prevent Aliasing bandlimit the input signals.

Band-limit is accomplished by using analog low-pass filters that are called anti aliasing filters.

(c) Explain flat top sampling technique.

During transmission, noise is introduced at top of the transmission pulse which can be easily removed if the pulse is in the form of flat top. Here the top of the sample is flat; they have constant amplitude.

It is also known as practical sampling.

- (d) Find the Nyquist rate and Nyquist interval for the signal $x(t) = \cos(4000\pi t) \cos(1000\pi t)$. CSE214059

We know, $\cos A \cos B = \frac{1}{2} [\cos(A+B) + \cos(A-B)]$

$$x(t) = \frac{1}{2} [\cos 5000\pi t + \cos 3000\pi t]$$

So new equation,

$$x(t) = \frac{1}{2} [\cos \omega_1 t + \cos \omega_2 t]$$

$$\omega_1 = 5000\pi$$

$$f_1 = \frac{5000\pi}{2\pi} = 2500 \text{ Hz}$$

$$f_m = 2500 \text{ Hz}$$

$$\omega_2 = 3000\pi$$

$$f_2 = \frac{3000\pi}{2\pi} = 1500 \text{ Hz}$$

- Nyquist rate is given as,

$$f_s = 2f_m$$

$$= 2 \times 2500 = 5000 \text{ Hz}$$

$$f_s = 5000 \text{ Hz}$$

- Nyquist interval $(T) = \frac{1}{2f_m} = \frac{1}{f_s} = \frac{1}{5000} = 0.2 \text{ ms}$

$$T = 0.2 \text{ ms}$$

SOLUTION - 5

- (a) How is an analog signal converted to digital signal in PCM system?

An encoder is a device that converts the analog signal to digital pulses. It response to each sample by generating a binary pulse or pattern. The combination of low pass filter, quantizer and encoder works as an A/D or Analog to digital converter.

Q) Derive the expression for transmission bandwidth in a PCM system. CSF21405.9

TRANSMISSION BANDWIDTH,

$V = \text{no. of binary digit to represent each line}$

$q = 2V = \text{no. of bits}$

No. of bits per sample = V

No. of sample per second = f_s

bit - rate = no. of bits per sample $\times$ no. of bits per sec

$$r = V \times f_s$$

$$BW \geq \frac{1}{2} r = \frac{1}{2} V f_s = \frac{1}{2} \times 2 f_m V$$

$$\therefore \text{Bandwidth (BW)} = V f_m$$

(c) A television signal having a bandwidth of 4.2 MHz is transmitted using binary PCM system. Given that the no. of quantisation level is 512. Determine transmission bandwidth and output signal to quantisation noise ratio.

Given ~~fm~~ $f_m = 4.2 \text{ MHz}$ and quan. level = 512.

$$\text{code word} : 2^n = 512$$

$$\therefore n \rightarrow \text{no. of bits} = 9$$

We know that transmission channel BW $\geq 0.5 n f_m$

$$\therefore BW \geq 0.5 \times 9 \times 4.2 \times 10^6 \text{ Hz}$$

$$BW \geq 37.8 \text{ MHz}$$

Output signal to quantisation noise ratio,

$$\therefore \left(\frac{S}{N} \right) \leq (4.8 + 6n) \text{ dB}$$

$$\Rightarrow \left(\frac{S}{N} \right) \leq (4.8 + 6 \times 9) \text{ dB} = \left(\frac{S}{N} \right) \leq 58.8 \text{ dB (ans)}$$